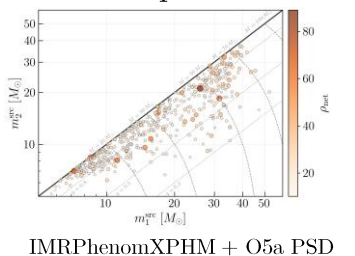


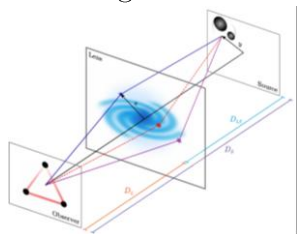
## Stage 1 (Ch. 5): Population Generation

Event  
Generation

Source &  
Lens Population



Lensing Simulation



EPL + External Shear + Thin  
Lens + Geometric Optics

Detectability filter

152k candidates drawn  
↓  
500 detectable ( $\rho_{\text{net}} \geq 10$ )  
↓  
309 pass duty cycle (0.70)  
↓  
157 final events  
( $\geq 2$  img,  $\rho_{\text{net}} \geq 10$ )

## Stage 2 (Ch. 6): Pipeline setup

Templates

- Single-image and joint *.ini* templates + SLURM script templates

*Lenskyloc\_inj\_hanabi\_single.ini*

*Lenskyloc\_inj\_hanabi\_joint.ini*

File generation

*Generator\_inis\_lenskyloc.py* populates priors and injection params per event

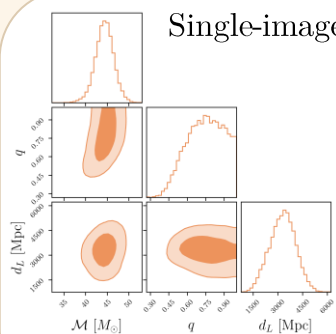
Output files

Two single-image *.ini* files and one joint *.ini* file per event + SLURM scripts for post-processing (result merging, SNR recovery, sky map generation)

## Stage 3 (Ch. 6): Parameter estimation

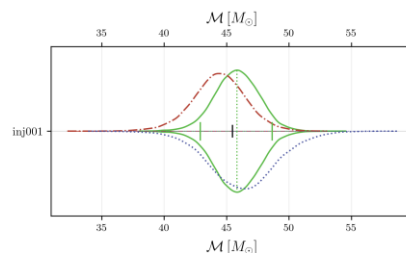
**BILBY**

Single-image PE



Each image analyzed separately

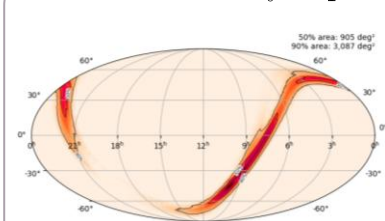
Joint PE



Coherent pair analysis: shared sky  
location, independent distances

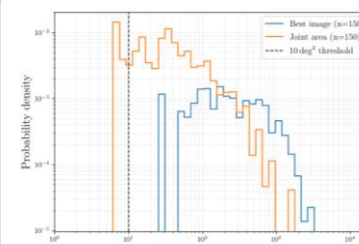
## Stage 4 (Ch. 7): Sky localization

Posterior to skymap



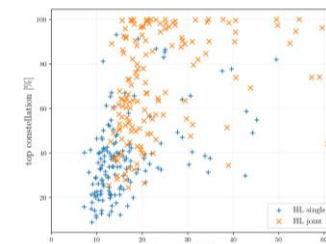
Sky maps from posterior samples

90% credible area



Extract sky area per analysis

Single vs joint



Extract sky area per analysis